SUNWAY UNIVERSITY

Jeffrey Cheah Foundation
Nurturing the Seeds of Wisdom

Topic 01: Introduction to Distributed Systems

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NET3204
Distributed Systems

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Learning Outcome

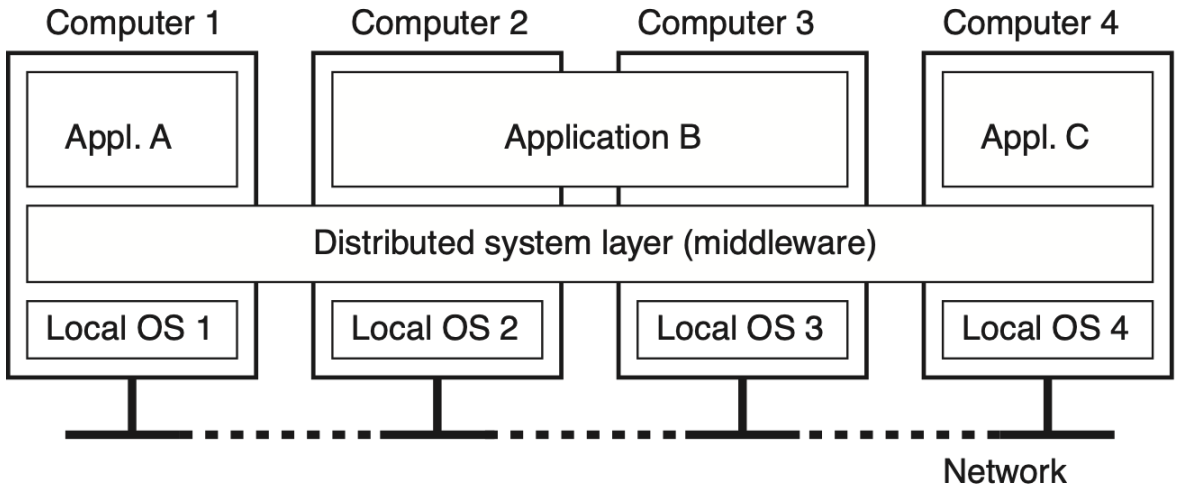
- At the end of this lecture, you will be able to:
  - Understand the concept of distributed systems.
  - Explain the goal and design challenges of implementing distributed systems.

What is a Distributed System?

- A system in which components located at networked computers communicate and coordinate their actions by message-passing.
- A distributed system consists of a collection of autonomous computers linked by a network that appears to the users of the system as a single computer. It is equipped with software that enables computers to coordinate their activities and to share resources (h/w, s/w, and data)
- “several computers doing something together”

Two aspects of distributed systems

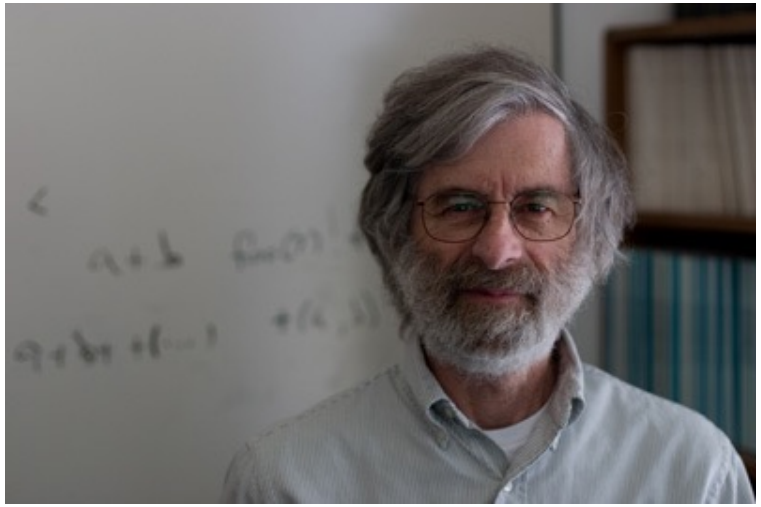
- Independent computers
  - e.g. desktop, laptop, handheld devices, sensors
- Single system
  - achieved using middleware



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What Leslie Lamport said?

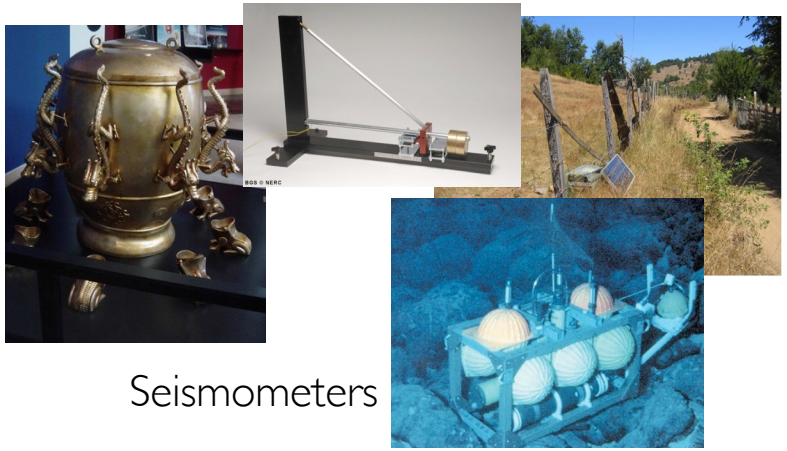
- "A distributed system is one on which I cannot get any work done because some machine I have never heard of has crashed.“
  - Leslie Lamport – a famous researcher on timing, message ordering, and clock synchronisation in distributed systems.
- Examples
  - World Wide Web
  - Credit card transaction systems
  - Some of your mobile apps



Scientific Gateway for Seismologists



EarthQuake Data Portal
www.seismicportal.eu



Seismometers

gathered



Seismic data archives

used



HPC Resources in LRZ: 147,456 cores,
3.185 PFlop/s, 44PB storage

processed

Source: Virtual Earthquake and seismology Research Community e-science environment in Europe
www.verce.eu

Characteristics of distributed systems

- Concurrency of components
  - Coordination of concurrently executing programs that share resource
- Lack of global clock
  - Coordination action through message exchange, rely on shared idea of time, how???
- Independent failures of components
  - Each component can fail independently, leaving others running.

Driving factors for Distributed Systems

- Ever increasing need to SHARE resources and information
- Widely use of sensor and mobile instruments
- Rapid decreasing cost of powerful workstations
- Widespread use of networks
- Maturity of software technology

- Your task : what are the other factors???

Goals of Distributed Systems

- Efficiency
  - Communication
  - Load Distribution
- Flexibility
  - Users viewpoint:
    - ease of use, consistency, reliability
  - Systems viewpoint:
    - modularity, scalability, portability, interoperability

Goals of Distributed Systems

- Consistency
  - A system is consistent if there is uniformity in using the system and the system behaviour is predictable.
  - Integrity of the system must be maintained with proper concurrency control
  - Inconsistency can be due to lack of global info, potential replication and partitioning of data, possibility of component failures or complexity of interaction among modules
- Robustness
  - Able to handle exceptional situations and errors, ex: changes in topology, long message delays, inability to locate servers.
  - Should include security, reliability, protection and access control







LOFAR stations from Google Maps

- 18 Core station fields (96 LBA, 2 x 24 HBA Tiles) < 2 km
- 18 Remote station fields (96 LBA, 48 HBA Tiles) < 100 km
- 8 International station fields (96 LBA, 96 HBA Tiles) < 1500 km
- low band antennas generate 1.6 TB raw data per second
- data processing pipeline: correlating and reducing data, real-time analysis and model tuning, storing the raw data and archiving the final data products

Source: LOFAR -- Low-Frequency Array for radio astronomy http://www.lofar.org/

Challenges in designing a Distributed System

- Heterogeneity (variety and difference)
- Openness
- Security
- Scalability
- Failure handling
- Concurrency
- Transparency

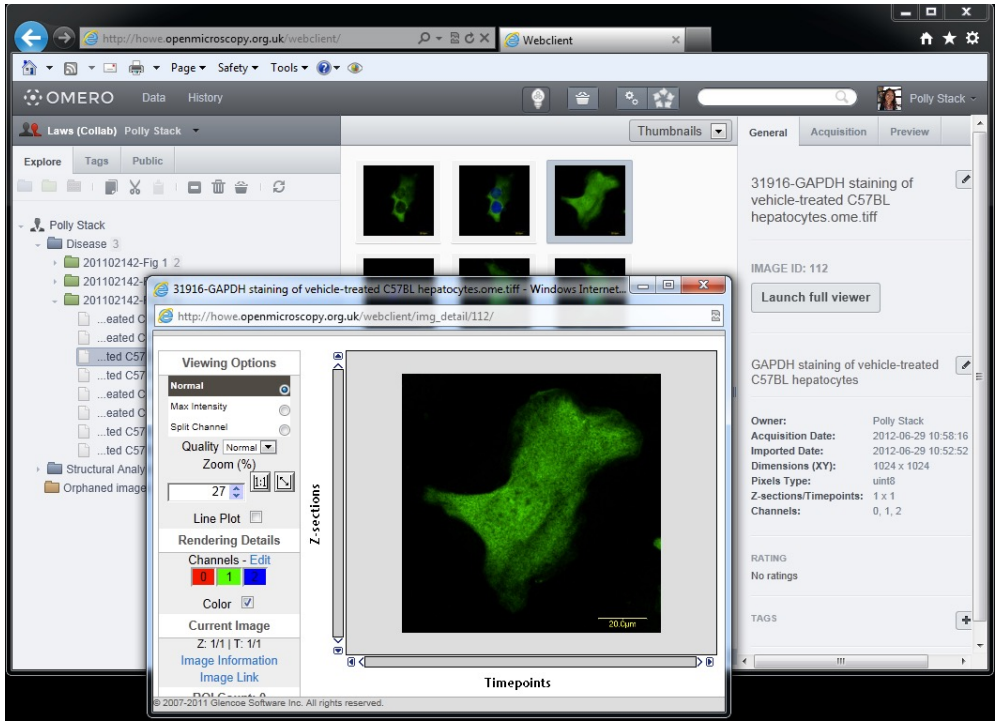
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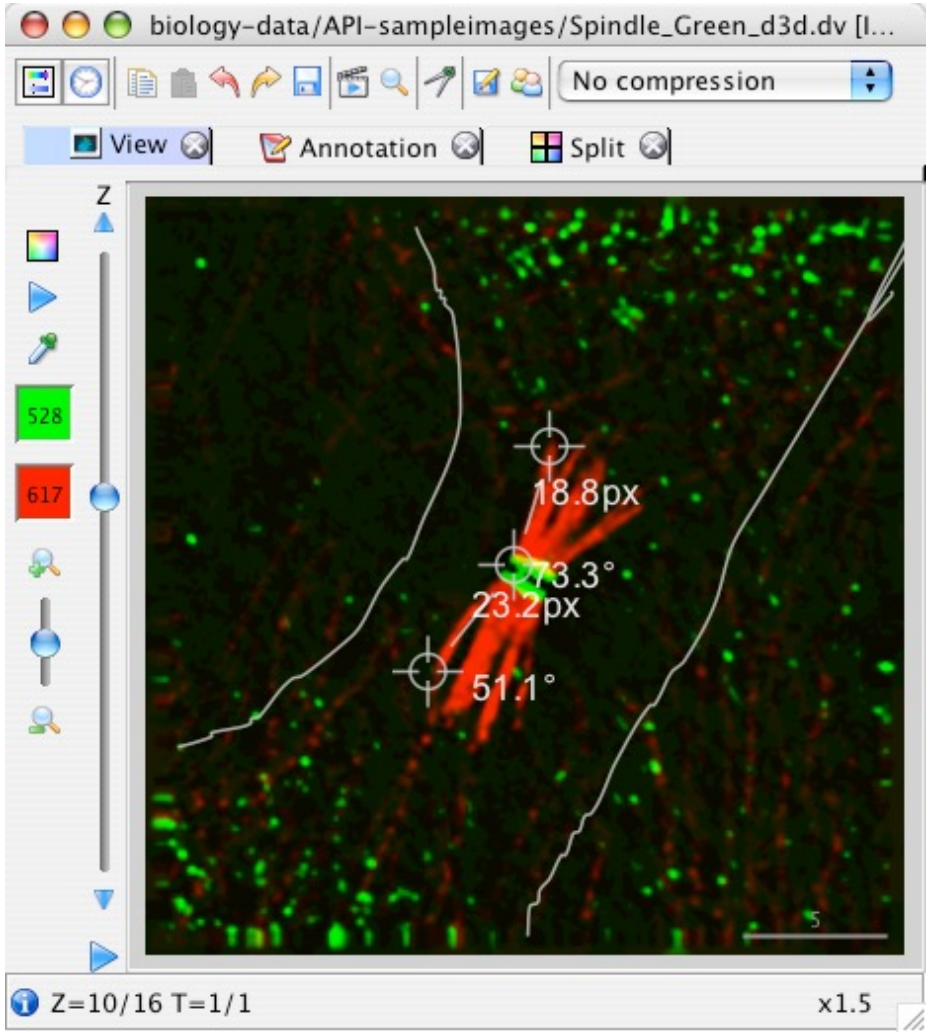
Heterogeneity

- Networks
- computer h/w
- OS
- Programming Languages
- Implementations by different developers
- Data formats
- Execution models

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Data management for experimental biology





- >100 file formats
- size range from KB to GB

Source: Open Microscopy Environment Remote Objects (OMERO) http://www.openmicroscopy.org

Openness

- the degree to which new resource-sharing services can be added and be made available for use by a variety of client program
- Can be extended at
  - the h/w level
  - the s/w level
- Be able to interact with services from other open systems, irrespective of the underlying environment:
  - Systems should conform to well-defined interfaces
  - Systems should support portability of applications
  - Systems should easily interoperate

Security

- concealing the contents of message.
- knowing for sure the identity of the agent correctly.
- identify a remote user or other agent correctly.

- Solution – encryption techniques.
- Three components of security:
  - Confidentiality
  - Integrity
  - Availability

Scalability

- A system is described as scalable if will remain effective when there is a significant increase in the number of resources and the number of users. eg. Internet.
- At least three components:
  - Number of users and/or processes (size scalability)
  - Maximum distance between nodes (geographical scalability)
  - Number of administrative domains (administrative scalability)

Techniques for Scaling

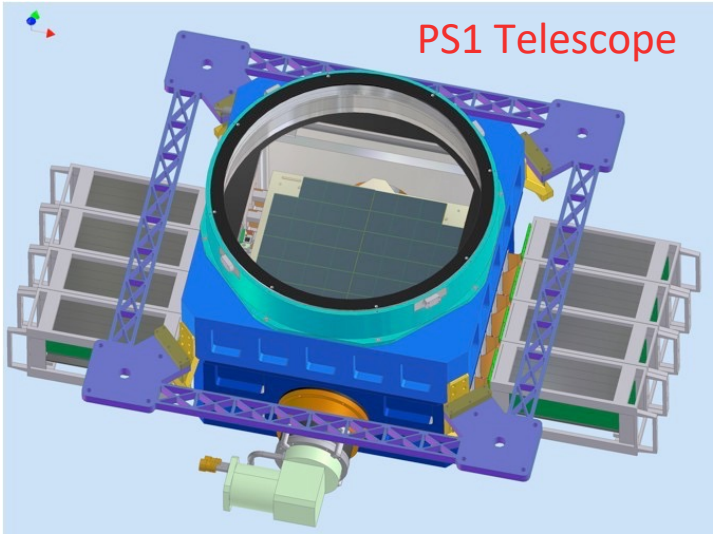
- Hide communication latencies
  - Avoid waiting for responses; do something else:
    - Make use of asynchronous communication
    - Have separate handler for incoming response
- Distribution
  - Partition data and computations across multiple machines:
    - Move computations to clients (Java applets)
    - Decentralised naming services (DNS)
- Replication/caching
  - Make copies of data available at different machines:
    - Mirrored Web sites
    - Web caches (in browsers and proxies)

Failure handling

- Computer systems sometimes fail. When fault occur in h/w or s/w, programs may produce incorrect results or they may stop before they have completed the intended computation.
- Techniques for dealing with failure:
  - Detecting failure – use checksums.
  - Masking failures – hidden the failure (eg. retransmission, backup storage).
  - Tolerating failures.
  - Recovery from failures.
  - Redundancy.

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- 4 x 1.4 Gigapixel resolution digital cameras
- creates ~700 new Load databases storing nightly detected object (each day)
- merges 50,000 Load databases with existing 12 offline cold databases (each week).
- capture > 1 PB of raw data and generate 100 TB data into the catalogue database (each year).



Source: Pan-STARRS -- the Panoramic Survey Telescope & Rapid Response System http://pan-starrs.ifa.hawaii.edu

Concurrency

- allows clients to access a share resources at the same time.
  - eg. accessing a data string that records bids for an auction.
- Any object (services/applications) that represents a shared resource in a DS must be responsible for ensuring that it operates correctly in a concurrent environment.

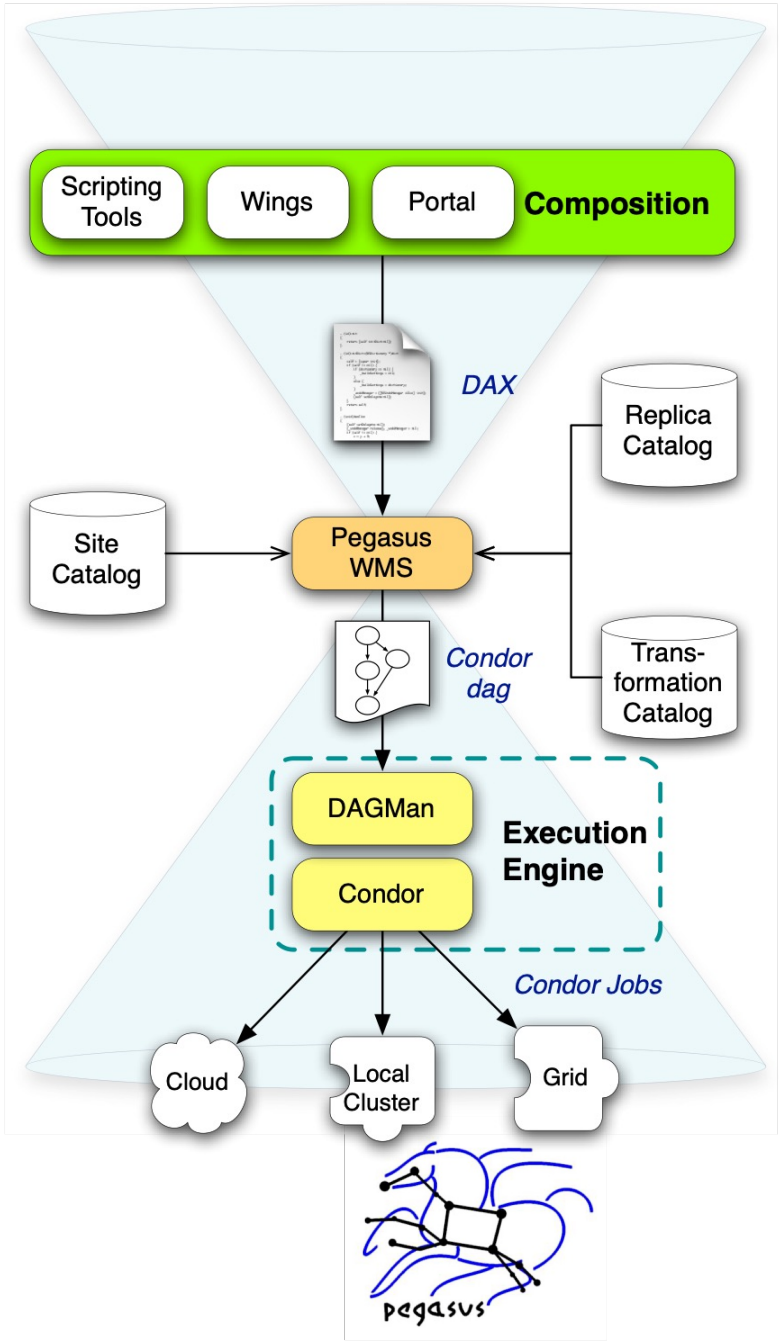
Transparency

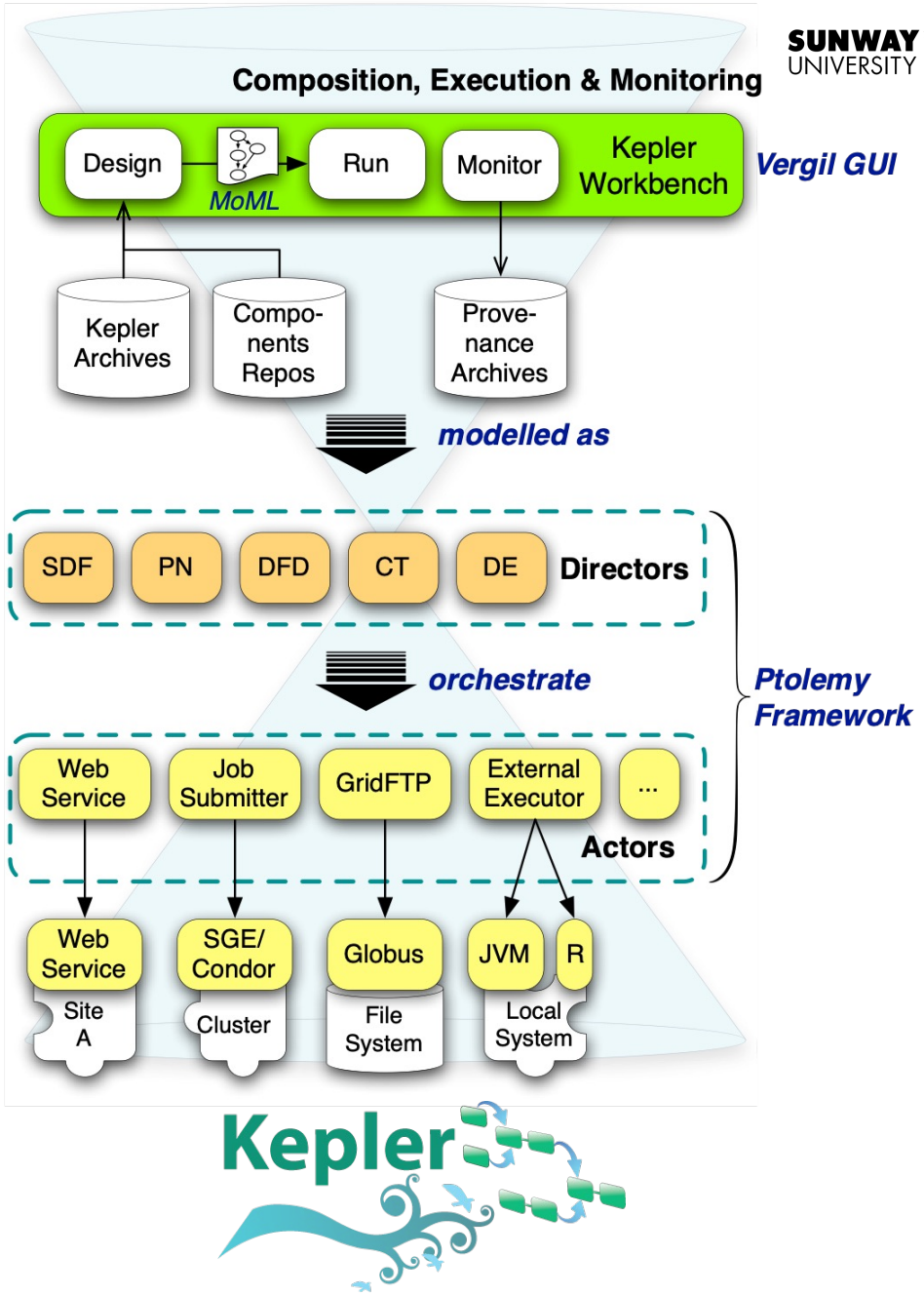
- the concealment from the user and the application programmer of the separation of components in a distributed system,
  - so that the system is perceived as a whole rather than as a collection of independent components.

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Types of transparency

- Access
  - Hides differences in data representation and invocation mechanisms
- Location
  - Hides where an object resides
- Migration
  - Hides from an object the ability of a system to change that object's location
- Relocation
  - Hides from a client the ability of a system to change the location of an object to which the client is bound
- Replication
  - Hides the fact that an object or its state may be replicated and that replicas reside at different locations
- Concurrency
  - Hides the coordination of activities between objects to achieve consistency at a higher level
- Failure
  - Hides failure and possible recovery of objects

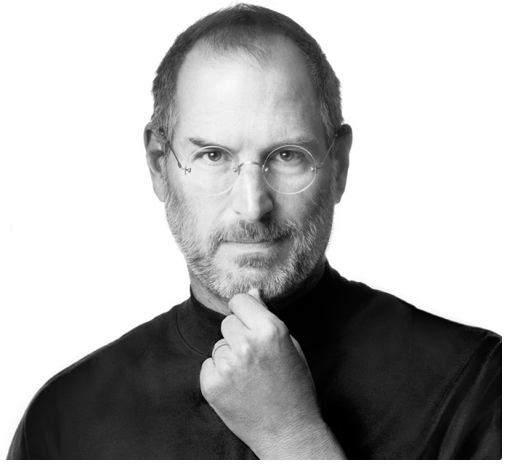

Scripting Tools
Wings
Portal
Composition
DAX
Replica Catalog
Site Catalog
Pegasus WMS
Condor dag
Trans-formation Catalog
DAGMan
Condor
Execution Engine
Condor Jobs
Cloud
Local Cluster
Grid
pegasus


Composition, Execution & Monitoring
Design
MoML
Run
Monitor
Kepler Workbench
Vergil GUI
Kepler Archives
Compo-nents Repos
Prove-nance Archives
modelled as
SDF
PN
DFD
CT
DE
Directors
orchestrate
Ptolemy Framework
Web Service
Job Submitter
GridFTP
External Executor
...
Actors
Web Service
Site A
SGE/ Condor
Cluster
Globus
File System
JVM
R
Local System
Kepler

Developing distributed system: Pitfalls

- False assumptions made:
  - The network is reliable
  - The network is secure
  - The network is homogeneous
  - The topology does not change
  - Latency is zero
  - Bandwidth is infinite
  - Transport cost is zero
  - There is one administrator

That's been one of my mantras – **focus** and **simplicity**. Simple can be harder than complex: you have to work hard to get your thinking clean to make it simple.



Steve Jobs

Summary

- What did I learn in this module?
  - The concept of distributed systems
  - The characteristics of distributed systems
  - The goals of distributed systems
  - The challenges in implementing distributed systems

Acknowledgement

The figures and contents used in this lecture are adopted and adapted from:

- van Steen, M. and Tanenbaum A.S. (2023). Distributed Systems, 4th Edition. distributed-systems.net.
- Colouries, G., Dollimore, J., Kindberg, T and Blair, G. (2021). Distributed Systems: Concepts and Design, 5th Edition. Pearson.
- Web references.

THANK YOU

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